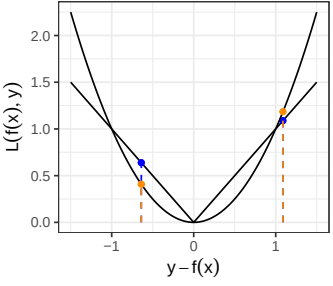
# Introduction to Machine Learning

# Supervised Regression: Linear Models with $L1$ Loss

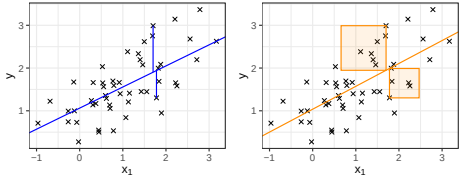



**Learning goals**

- Understand difference between $L1$ and $L2$ regression
- See how choice of loss affects optimization & robustness

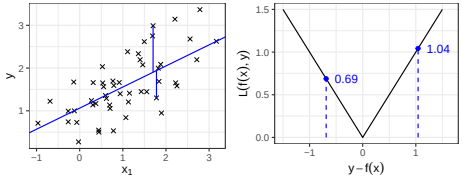
# ABSOLUTE LOSS

- $L2$ regression minimizes quadratic residuals – wouldn't **absolute** residuals seem more natural?

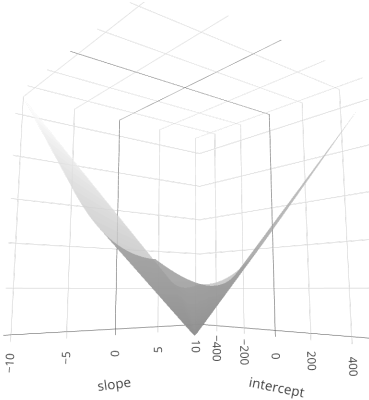


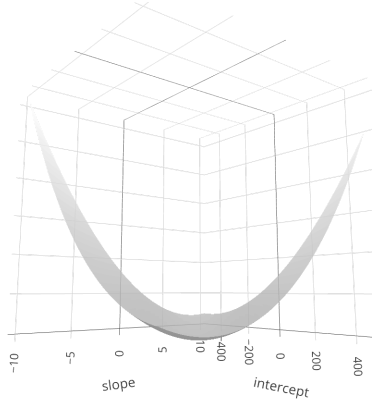

- $L1$ **loss / absolute error / least absolute deviation (LAD)**

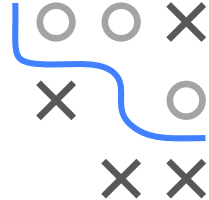
$$L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$$



# $L1$ VS $L2$ – LOSS SURFACE



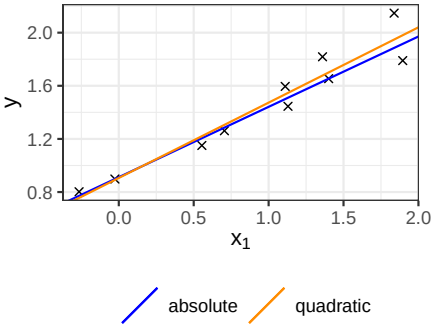
$L1$ loss (left) harder to optimize than $L2$ loss (right)

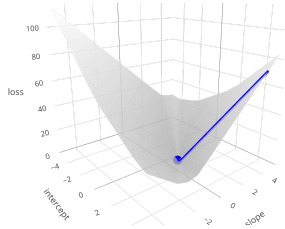
- Convex but **not differentiable** in $y - f(\mathbf{x}) = 0$
- No analytical solution

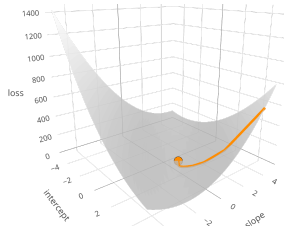
# $L1$ VS $L2$ – ESTIMATED PARAMETERS

- Results of $L1$ and $L2$ regression often not that different
- Simulated data: $y^{(i)} = 1 + 0.5x_1^{(i)} + \epsilon^{(i)}, \quad \epsilon^{(i)} \overset{i.i.d}{\sim} \mathcal{N}(0, 0.01)$

| | intercept | slope |
|---|---|---|
| $L1$ | 0.91 | 0.53 |
| $L2$ | 0.91 | 0.57 |

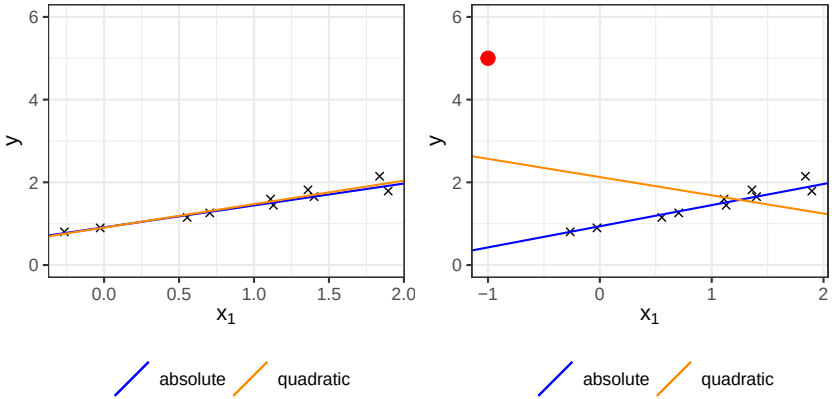


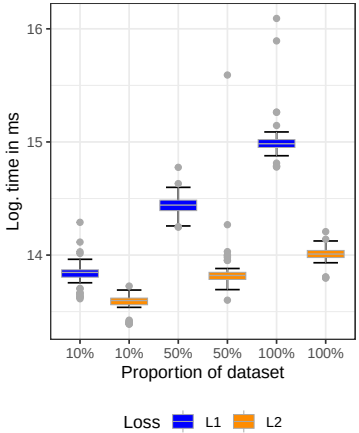
# $L1$ VS $L2$ – ROBUSTNESS

- $L2$ quadratic in residuals $\rightsquigarrow$ outlying points carry lots of weight
- E.g., $3\times$ residual $\Rightarrow$ $9\times$ loss contribution
- $L1$ more **robust** in presence of outliers (example ctd.):



# $L1$ VS $L2$ – OPTIMIZATION COST

- Real-world $\mathrm{weather}$ problem $\rightsquigarrow$ predict mean temperature
- Compare **time** to fit $L1$ ($\mathrm{quantreg::rq()}$) vs $L2$ ($\mathrm{lm::lm()}$) for different dataset proportions (repeat 50$\times$)




Loss

| | Fitted: $L1$ | Fitted: $L2$ |
|---|---|---|
| Total $L1$ loss | $8.98 \times 10^4$ | $8.99 \times 10^4$ |
| Total $L2$ loss | $5.83 \times 10^6$ | $5.81 \times 10^6$ |

Estimated coefficients

| $x_j$ | $L1$: $\hat{\theta}_j$ | $L2$: $\hat{\theta}_j$ |
|---|---|---|
| $\mathrm{Max_temperature}$ | 0.553 | 0.563 |
| $\mathrm{Min_temperature}$ | 0.441 | 0.427 |
| $\mathrm{Visibility}$ | 0.026 | 0.041 |
| $\mathrm{Wind_speed}$ | 0.002 | 0.010 |
| $\mathrm{Max_wind_speed}$ | $-0.026$ | $-0.039$ |
| $(\mathrm{Intercept})$ | $-0.380$ | $-0.102$ |

$L1$ **slower** to optimize!